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Arnab Mukherjee, Ankit Raj and Anuj Bangad

Centre for Machine Intelligence and Data Science,
IIT Bombay

Faculty: Prof. Jayakrishnan Nair and Prof. Sharayu Moharir

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Supervised vs Unsupervised Learning I

A customer analytics problem

- **Dataset Information (column headers):**
 - Customer age,
 - Annual income,
 - Spending score,
 - Number of purchases per month,
 - Premium membership status (Yes/No).
- Identify which task falls into **supervised** and **unsupervised**, and mention an **ML algorithm** to solve:
 - Task 1: Predict whether a new customer will purchase a premium membership,

Supervised vs Unsupervised Learning II

- Task 2: Group customers into different categories such as high spenders, budget shoppers, and occasional buyers.

Supervised vs Unsupervised Learning III

- Answer:
 - First task falls into **supervised learning** since **target variable (Premium membership) is already available**. **Logistic regression** can be used.
 - Second task falls into **unsupervised learning** since **target variable (group) is unavailable**. **K-means** can be used.

Convexity, Optimization and Linear Algebra I

Let

$$f(\theta_1, \theta_2) = \theta_1^2 + \theta_2^2 + \theta_1\theta_2$$

Check whether f is convex. If yes, find the global minima and the global minimum value directly.

Convexity, Optimization and Linear Algebra II

$$f(\theta_1, \theta_2) = \theta_1^2 + \theta_2^2 + \theta_1\theta_2$$

The gradient of f is

$$\nabla f(\theta_1, \theta_2) = (2\theta_1 + \theta_2, \theta_1 + 2\theta_2)$$

Setting $\nabla f = 0$,

$$2\theta_1 + \theta_2 = 0, \quad \theta_1 + 2\theta_2 = 0$$

Solving, we get critical points

Convexity, Optimization and Linear Algebra III

$$\theta_1 = 0, \quad \theta_2 = 0$$

Now compute the Hessian matrix:

$$\nabla^2 f(\theta_1, \theta_2) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Its leading principal minors are

$$2 > 0, \quad \det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = 3 > 0$$

Convexity, Optimization and Linear Algebra IV

Hence, the **Hessian is positive definite**, so f is **strictly convex**.

Therefore, **the critical point is the unique global minimizer**.

Thus, the global minimum occurs at

$$(\theta_1, \theta_2) = (0, 0)$$

with minimum value

$$f(0, 0) = 0$$

Statistical Framework, Hypothesis Testing I

Question:

Consider the following simple linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n$$

where y_i denotes the target variable and x_i is a feature.

Suppose the fitted regression model from $n = 5$ observations gives

$$\hat{\beta}_1 = 2, \quad SE(\hat{\beta}_1) = 0.5$$

Test whether the feature x has a significant effect on predicting y at the 5% significance level.

Statistical Framework, Hypothesis Testing II

Solution:

We test

$$H_0 : \beta_1 = 0$$

against

$$H_1 : \beta_1 \neq 0$$

The test statistic is

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

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Statistical Framework, Hypothesis Testing III

Substituting the values,

$$t = \frac{2}{0.5} = 4$$

The degrees of freedom are

$$n - 2 = 5 - 2 = 3$$

At the 5% significance level, the critical value is

$$t_{0.025,3} \approx 3.182$$

Since

Statistical Framework, Hypothesis Testing IV

$$|4| > 3.182$$

we reject H_0 .

Hence, there is sufficient evidence to conclude that the feature x has a significant effect on predicting the target variable y .

Gradient Descent I

Given:

$$f(x, y) = x^2 + y^2 + xy$$

Use gradient descent with:

$$\eta = 0.2$$

Starting point:

$$(1, 2)$$

Find:

- Gradient vector
- One iteration
- Whether the objective decreases

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Gradient Descent II

Gradient:

$$\nabla f(x, y) = \begin{bmatrix} 2x + y \\ 2y + x \end{bmatrix}$$

At (1, 2):

$$\nabla f(1, 2) = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Update:

$$x_1 = 1 - 0.2(4) = 0.2$$

$$y_1 = 2 - 0.2(5) = 1$$

Gradient Descent III

New point:

$$(0.2, 1)$$

Function values:

$$f(1, 2) = 1^2 + 2^2 + (1)(2) = 7$$

$$f(0.2, 1) = (0.2)^2 + 1^2 + (0.2)(1) = 1.24$$

Since:

$$1.24 < 7$$

Objective decreases.

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Convex Optimization I

Solve:

$$\min_{x,y} x^2 + y^2$$

Subject to:

$$x + y = 1$$

Find:

- Lagrangian
- Stationary point
- Optimal value

Convex Optimization – Solution (Part 1) I

Lagrangian:

$$L(x, y, \lambda) = x^2 + y^2 + \lambda(x + y - 1)$$

Stationarity conditions:

$$\frac{\partial L}{\partial x} = 2x + \lambda = 0$$

$$\frac{\partial L}{\partial y} = 2y + \lambda = 0$$

Therefore:

$$x = y$$

Convex Optimization – Solution (Part 2) I

Constraint:

$$x + y = 1$$

Substitute $x = y$:

$$x + x = 1$$

$$2x = 1$$

$$x = \frac{1}{2}$$

Convex Optimization – Solution (Part 2) II

Hence:

$$y = \frac{1}{2}$$

Objective value:

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Minimum value:

$$\boxed{\frac{1}{2}}$$

Primal Problem I

Solve:

$$\max \quad 3x + 2y$$

Subject to:

$$x + y \leq 4$$

$$x \geq 0, \quad y \geq 0$$

Use graphical method.

Primal Problem – Solution I

Corner points of the feasible region:

$$(0, 0), \quad (4, 0), \quad (0, 4)$$

Objective values:

$$f(0, 0) = 3(0) + 2(0) = 0$$

$$f(4, 0) = 3(4) + 2(0) = 12$$

$$f(0, 4) = 3(0) + 2(4) = 8$$

Primal Problem – Solution II

Maximum occurs at:

$(4, 0)$

Optimal value:

12

KKT Conditions I

Solve using KKT conditions:

$$\min_x x^2$$

subject to

$$x \geq 1$$

Tasks:

- Form Lagrangian
- Write KKT conditions
- Find optimal x and multiplier

KKT Conditions – Solution I

Lagrangian:

$$L(x, \lambda) = x^2 + \lambda(1 - x)$$

Stationarity:

$$2x - \lambda = 0$$

Complementary slackness:

$$\lambda(1 - x) = 0$$

At optimum:

$$x = 1$$

KKT Conditions – Solution II

Substitute:

$$2(1) - \lambda = 0$$

$$\lambda = 2$$

Optimal solution:

$$x^* = 1$$

Minimum value:

$$1$$

Model Assessment and Selection I

Question:

You have trained three different models on the same dataset:

- Model A: Training MSE = 15.2, Test MSE = 18.7
- Model B: Training MSE = 8.5, Test MSE = 25.3
- Model C: Training MSE = 12.1, Test MSE = 13.4

Which model would you select and why? Explain in terms of the bias-variance tradeoff.

Model Assessment and Selection II

Solution:

Analysis of each model:

- **Model A:**

$$\text{Gap} = 18.7 - 15.2 = 3.5$$

Moderate variance.

- **Model B:**

$$\text{Gap} = 25.3 - 8.5 = 16.8$$

High variance and overfitting.

- Very low training error
- High test error

Model Assessment and Selection III

- **Model C:**

$$\text{Gap} = 13.4 - 12.1 = 1.3$$

Low variance and good generalization.

Model Assessment and Selection IV

Conclusion:

Select **Model C** because:

- It has the lowest test MSE:

13.4

- It shows good generalization with a small gap between training and test errors.
- Model B suffers from overfitting (high variance), while Model C achieves the best bias-variance tradeoff.

Bootstrapping I

Question:

You have a sample of size:

$$n = 100$$

You perform bootstrap resampling with:

$$B = 1000$$

bootstrap samples.

- For a particular observation x_i in the original sample, what is the probability that it does NOT appear in a given bootstrap sample?
- Approximately what fraction of the original observations will be absent from any single bootstrap sample?

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Bootstrapping II

Solution:

In each draw, probability of NOT selecting x_i :

$$\frac{n-1}{n}$$

Since we draw n times with replacement:

$$P(x_i \text{ not in bootstrap sample}) = \left(\frac{n-1}{n}\right)^n$$

For $n = 100$:

$$P(x_i \text{ not in sample}) = \left(\frac{99}{100}\right)^{100} \approx 0.366$$

Bootstrapping III

General Result:

As:

$$n \rightarrow \infty$$

we have:

$$\left(\frac{n-1}{n}\right)^n \rightarrow e^{-1}$$

$$e^{-1} \approx 0.368$$

Therefore, approximately:

36.8%

of observations are absent from any given bootstrap sample.

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Bootstrapping IV

- Each bootstrap sample contains about:

63.2%

unique observations.

- Some observations appear multiple times.

Bootstrapping V

Implication for Model Validation:

- The observations not included in the bootstrap sample ($\approx 36.8\%$) can be used as a validation set.
- This forms the basis of the out-of-bag (OOB) error estimate used in Random Forests.
- With:

$$B = 1000$$

bootstrap samples, we obtain robust estimates of:

- Model performance
- Parameter uncertainty

In-Sample Methods I

Question:

A linear regression model is fit using 4 predictors and 50 observations. The training MSE of the model is 25. Assume that the error variance is $\sigma^2 = 30$.

Calculate the following in-sample model selection criteria:

- Mallow's C_p
- AIC
- BIC

In-Sample Methods II

Solution:

Mallow's C_p :

$$C_p = \frac{RSS}{\sigma^2} - n + 2(k + 1)$$

Residual Sum of Squares:

$$RSS = n \times MSE$$

$$RSS = 50 \times 25 = 1250$$

Substitute values:

$$C_p = \frac{1250}{30} - 50 + 2(5)$$

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In-Sample Methods III

$$C_p = 41.67 - 50 + 10$$

$$C_p \approx 1.67$$

Note:

$$k + 1 = 5$$

includes the intercept term.

In-Sample Methods IV

AIC and BIC:

AIC:

$$AIC = n \ln \left(\frac{RSS}{n} \right) + 2(k + 1)$$

$$= 50 \ln(25) + 2(5)$$

$$= 50(3.219) + 10$$

$$= 160.9 + 10 = 170.9$$

In-Sample Methods V

BIC:

$$BIC = n \ln \left(\frac{RSS}{n} \right) + \ln(n)(k + 1)$$

$$= 50 \ln(25) + \ln(50)(5)$$

$$= 160.9 + 19.6 = 180.5$$

In-Sample Methods VI

Interpretation:



$$BIC = 180.5 > AIC = 170.9$$

because BIC applies a stronger penalty for model complexity:

$$\ln(n) \quad \text{vs} \quad 2$$

- For model selection, lower values of:

$$AIC, BIC, C_p$$

are preferred.

In-Sample Methods VII

- BIC tends to select simpler models than AIC, especially for large n .
- Mallows's:

$$C_p \approx k + 1$$

suggests appropriate model complexity and no significant overfitting.